Survey on Probabilistic Planning of Explanations for Robot Actions

Dear Participant,

Thank you for taking part in our research study aimed at understanding the satisfaction and clarity associated with probabilistic plans of explanations for robot actions. The purpose of this study is to assess participants' ability to grasp the explanations for actions and behaviors of a navigation robot. During the study, you will be presented with a plan of robot actions. Following that, you will be requested to respond to a short questionnaire.

The anticipated time commitment for participating in this study is approximately 10 minutes. Your involvement is instrumental in enriching our understanding of how individuals perceive and follow the explanations of actions plans of navigation robots.

Your contribution to this research is entirely voluntary, and we hold your privacy and confidentiality in the highest regard. The information you provide will be used exclusively for research purposes. The study can also be terminated at any point if you do not wish to continue without consequences.

Information and Declaration of Consent on Data Protection

In scientific studies, personal data about you will be collected. The storage, evaluation, and transfer of this study-related data are carried out in accordance with legal regulations and requires the following voluntary consent prior to participation in the study:

1. I agree that data collected in the course of this study may be recorded on questionnaires and electronic data carriers and processed without mentioning my name.
2. Furthermore, I agree that an authorized person (e.g.: the client, the university), who is obligated to secrecy, may inspect my collected personal data, as far as this is necessary for the review of the project. For this measure, I release the researcher from professional secrecy.
3. I understand that I have the right to obtain information (including a free copy) about the personal data concerning me and to request its correction and deletion.

If you have any further questions, please contact us at amar.halilovic@uni-ulm.de

Thank you for your time! This study should take at most 10 minutes.

* Indicates required question

1. I consent to participate in the study and to the use of my data as described above. *

Mark only one oval.

- Yes

Demographic and Personality Details

2. Age *

_____

3. Sex *

Mark only one oval.

- Female
- Male
- Prefer not to say
- Other: _____

4. Completed level of education *

Mark only one oval.

- High School
- Bachelor Degree
- Master Degree
- Doctorate
- Other: _____

5. Profession *

_____

6. Do you have previous experience with Robotics? *

Mark only one oval.

- Yes
- No

7. Do you have previous experience with Artificial Intelligence? *

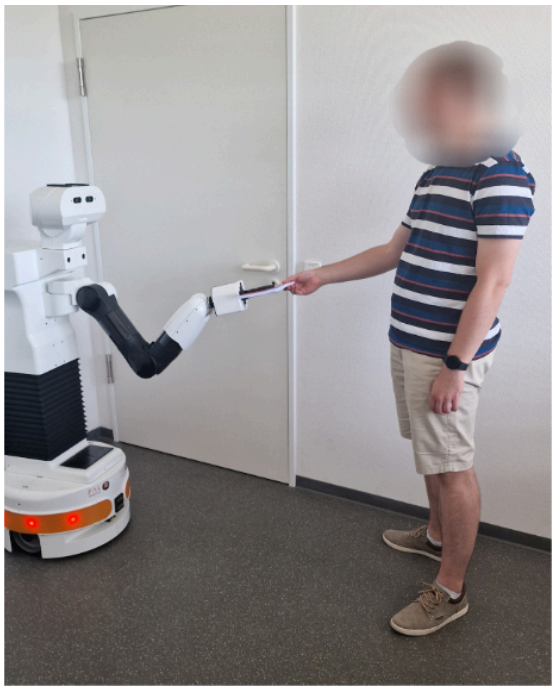
Mark only one oval.

- Yes
- No

Questionnaire

Scenario

Imagine a robot librarian with the task of fetching books for library visitors. Sometimes, fetching a book can come with unexpected problems, such as a delay because of an obstacle in the way, when the robot cannot find a requested book, or when the robot gets lost on the way. In such and similar cases, the robot should explain to the library visitors the reasons for the delay or return without a book. A robot can have two working modes, including explanations for humans and working silently. In this version, the robot is working silently. Please examine the robot's plan for serving two library visitors and answer the following questions.



Plan

1. **The robot is moving from the registration desk to the reading space 1.**
2. **The robot asks the visitor 1 for the title of the desired book.**
3. **The robot is moving to the bookshelf space.**
4. **The robot is fetching a book.**
5. **The robot is moving to the reading space 1.**
6. **Robot is rushing as it needs to serve an additional visitor.**
7. **The robot is giving the book to the visitor 1.**
8. **The robot is moving to the reading space 2.**
9. **The robot asks the visitor 2 for the title of the desired book.**
10. **The robot is moving to the bookshelf space.**
11. **The robot is fetching a book.**
12. **The robot is moving to the reading space 2.**
13. **The robot is delayed because of an obstacle in the way.**
14. **The robot is giving the book to the visitor 2.**
15. **The robot is going to the registration desk.**

8. **Curiosity checklist**

Imagine you are one of the visitors. Would you expect the robot to provide some explanations?

Check all that apply.

	Check all that apply
I want to know what the robot just did.	☐
I want to know that I understand this robot system correctly.	☐
I want to understand what the robot will do next.	☐
I want to know why the robot did not make some other decision.	☐
I want to know what the robot would have done if something had been different.	☐
I was surprised by the robot's actions and want to know what I missed.	☐

what I missed.

I would expect the robot to provide some explanations.

I would expect the robot to provide some explanations.

I find it helpful when robots explain their actions only when I ask questions.

I find it helpful when robots explain their actions only when I ask questions.

I find it helpful when robots describe and explain their actions without asking.

I find it helpful when robots describe and explain their actions without asking.

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